

Regression with a categorical predictor

Reminders and agenda

bring a calculator
(no phones)

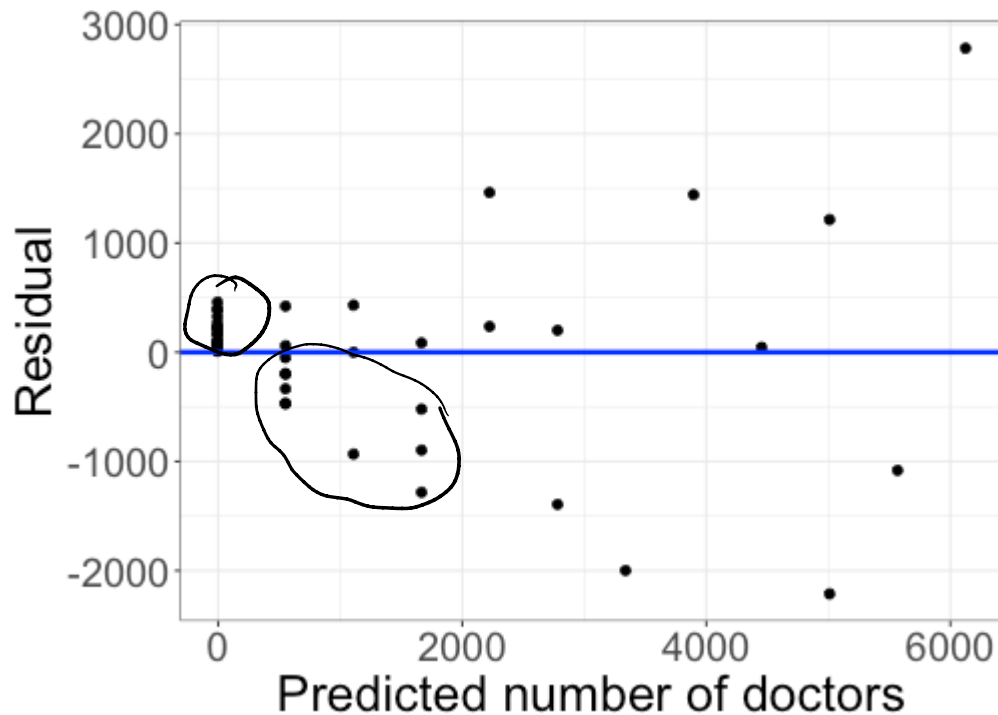
- + Exam 1 next Friday (10/10)
 - + Covers material through the last class (9/29)
- + Exam 1 review day next Wednesday (10/8)
- + Studying advice:
 - + Practice questions on the course website
 - + Review previous class activities
 - + Review previous homework assignments
- + Today: thinking about categorical explanatory variables

Last time: Class activity

Data on doctors and hospitals from a random sample of 53 US counties:

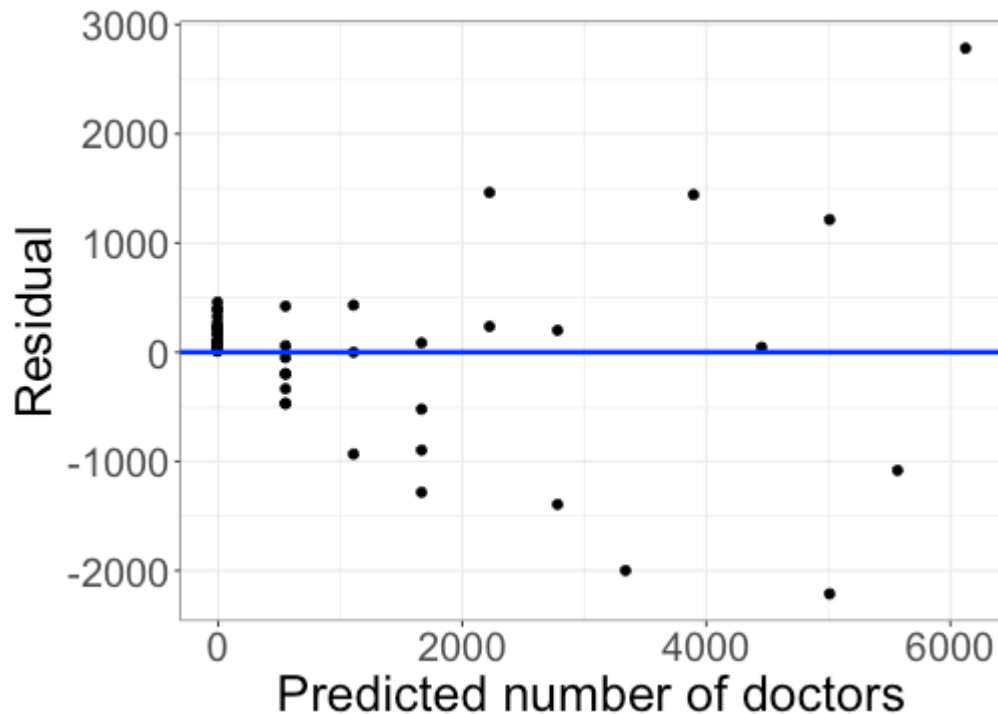
- + County: the name and state of the county
- + MDs: the number of medical doctors in the county
- + Hospitals: number of hospitals in the county
- + Beds: the number of beds in the hospital

Class activity



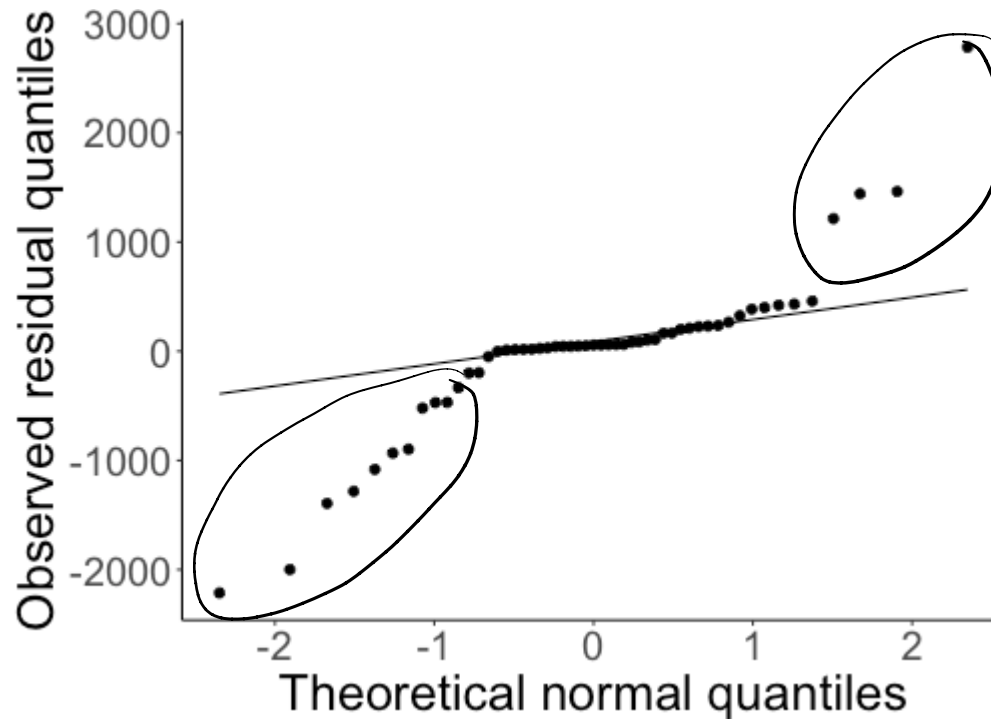
Do the shape and constant variance assumptions look reasonable?

Class activity



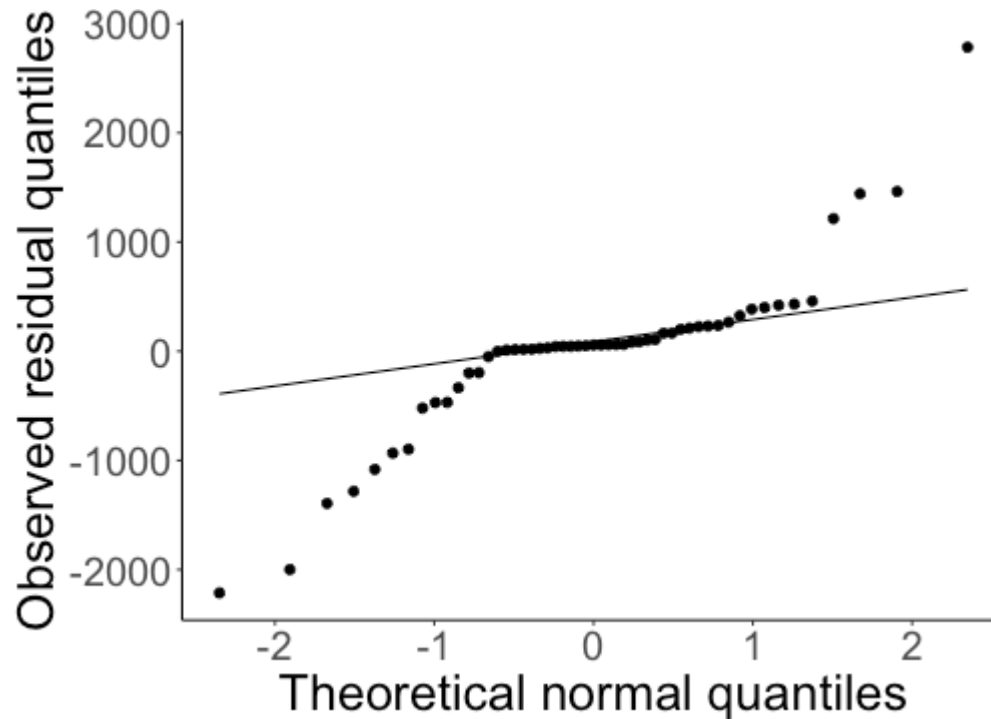
- + Shape may be slightly violated
- + Constant variance is definitely violated

Class activity



Does the normality assumption look reasonable?

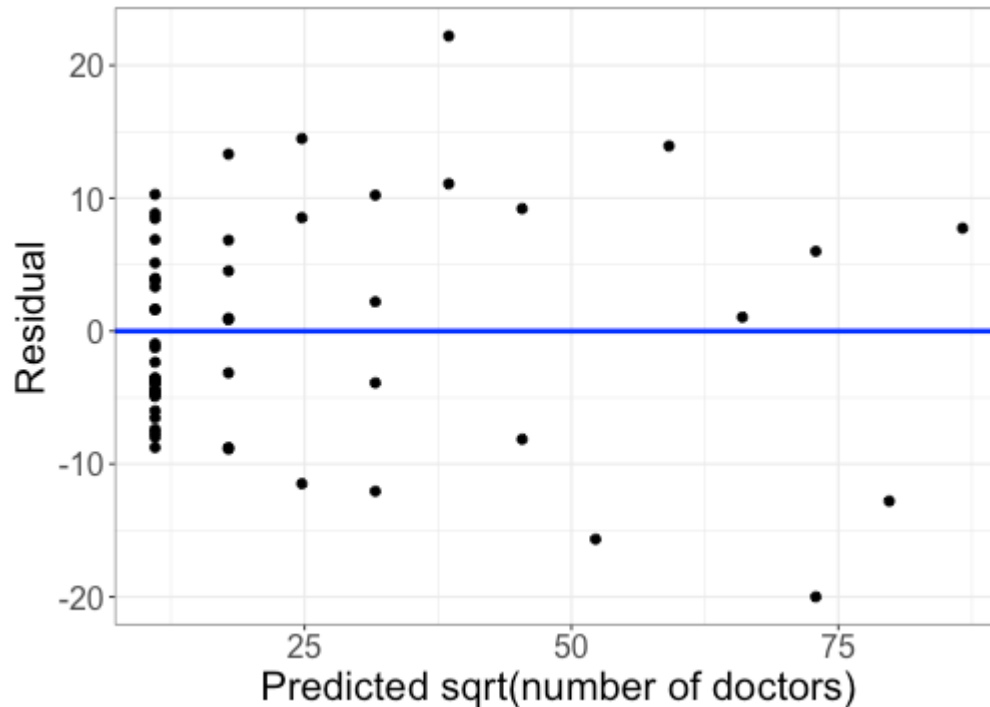
Class activity



The normality assumption is definitely violated.

Transformation

Trying a square root transformation of the response:

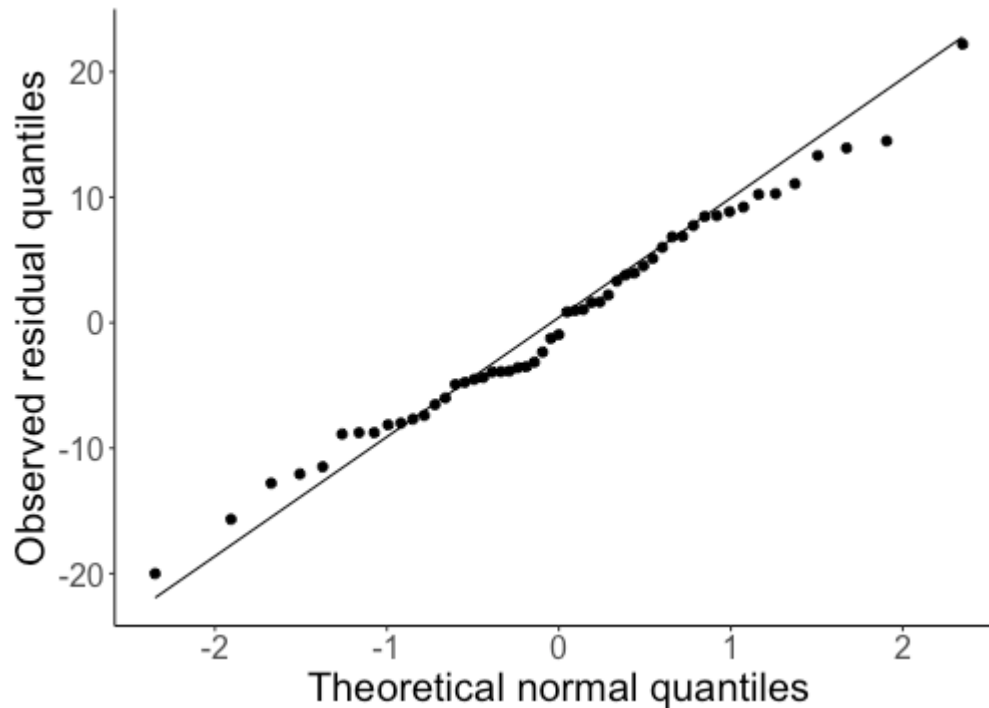


How do the shape and constant variance assumptions look now?

much better!

Transformation

QQ plot with the transformed data:



How does the normality assumption look now?

much better!

Class activity

Do the independence and randomness assumptions seem reasonable for the CountyHealth data?

Class activity

Do the independence and randomness assumptions seem reasonable for the CountyHealth data?

Yes, both seem reasonable.

- + Independence: reasonable to assume the number of hospitals/doctors for one county doesn't affect the number of hospitals/doctors for another county
- + Randomness: we are told the data come from a random sample

Regression so far

Simple linear regression:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

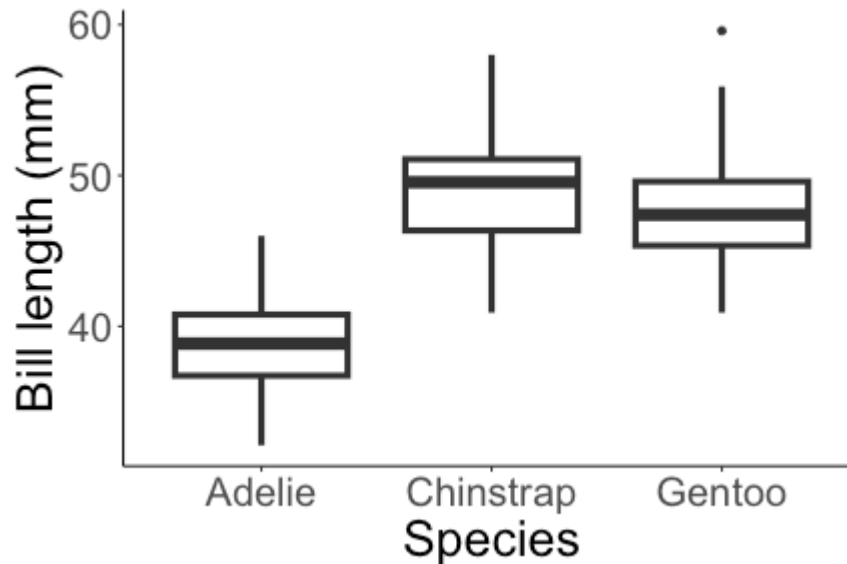
Polynomial regression:

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_k x^k + \varepsilon$$

where x and y are both **quantitative**.

What if x is *categorical*?

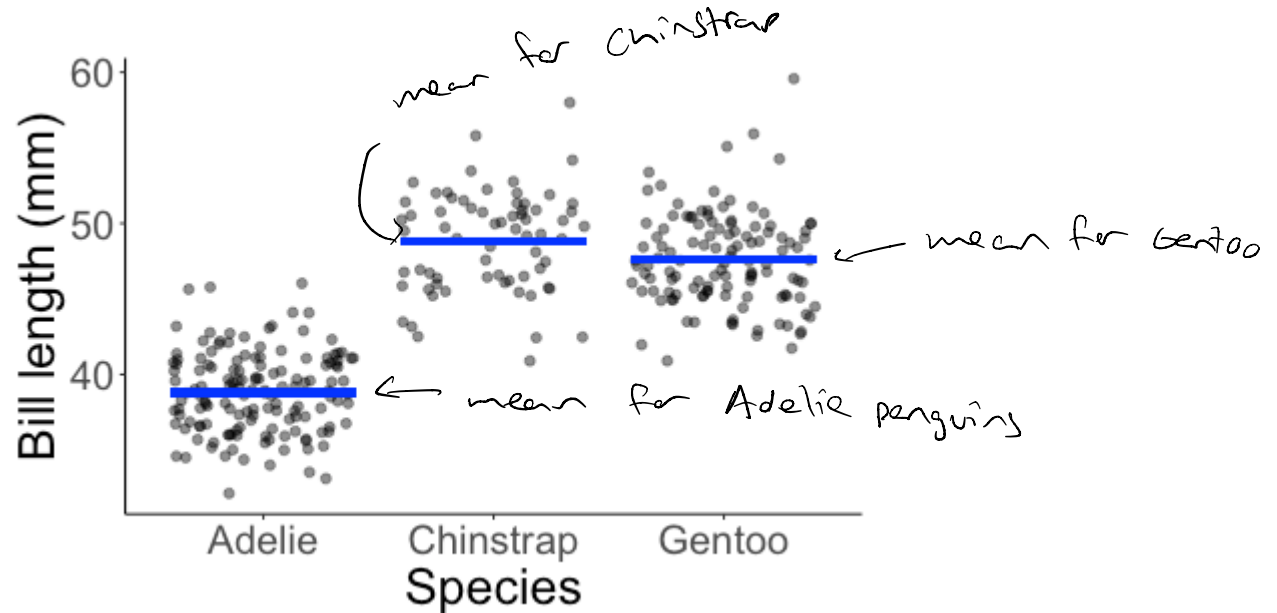
Categorical predictor



- + How do we model the relationship between species and bill length?
- + Can we test for a relationship between bill length and species?

Categorical predictor

Imagine population like this:

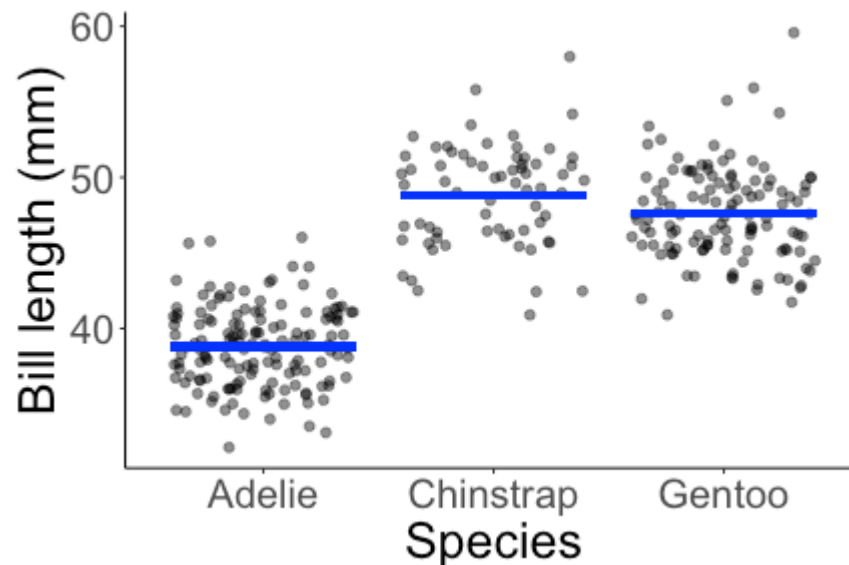


If species = Adelie,

$$\text{bill length} = \text{mean Adelie bill length} + \varepsilon$$

Categorical predictor

Imagine population like this:

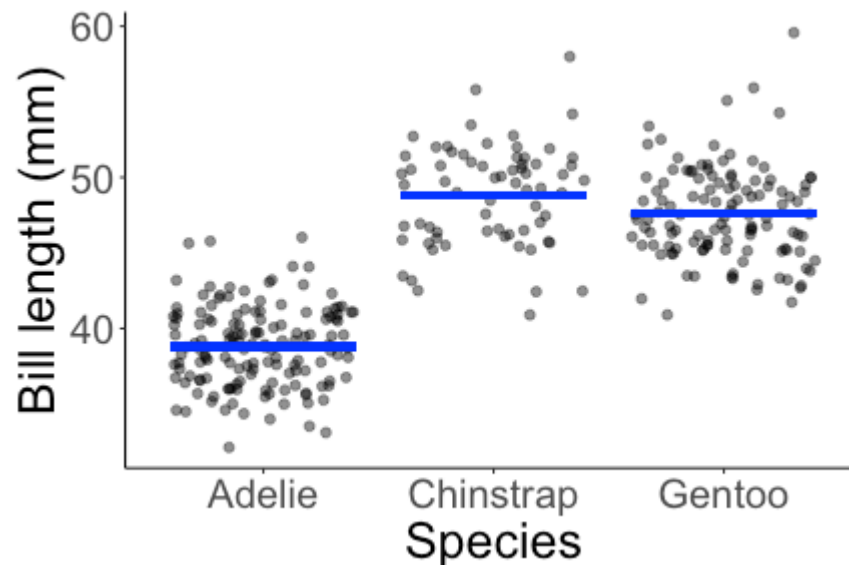


If species = Chinstrap,

$$\text{bill length} = \text{mean Chinstrap bill length} + \varepsilon$$

Categorical predictor

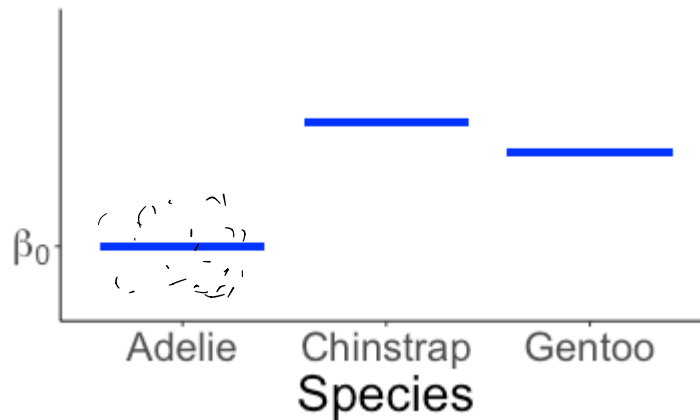
Imagine population like this:



If species = Gentoo,

$$\text{bill length} = \text{mean Gentoo bill length} + \varepsilon$$

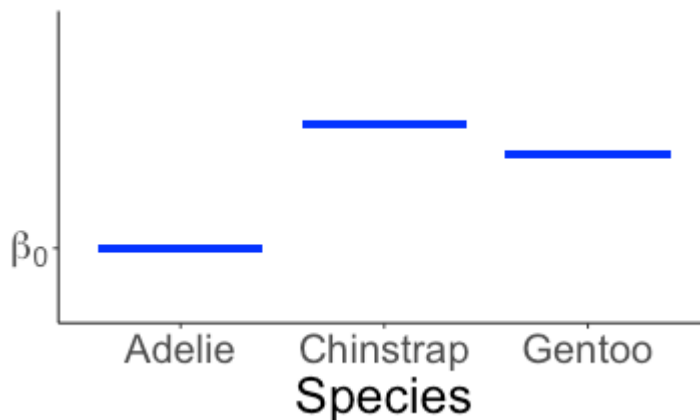
Categorical predictor



If species = Adelie, bill length = $\beta_0 + \varepsilon$ *← variability in the population*

+ β_0 = true mean bill length of Adelie penguins (in the population)

Categorical predictor

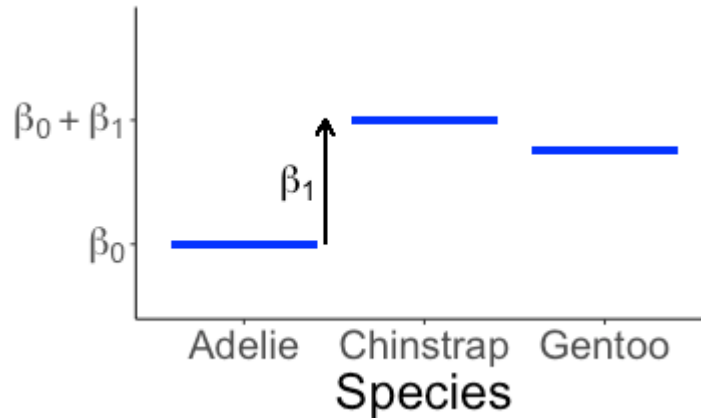


If species = Adelie, bill length = $\beta_0 + \varepsilon$

+ β_0 = true mean bill length of Adelie penguins (in the population)

What notation could I use for Chinstrap penguins?

Categorical predictor



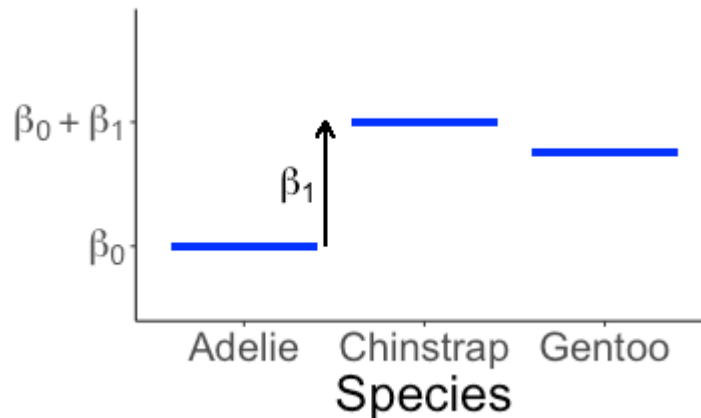
If species = Adelie, bill length = $\beta_0 + \varepsilon$

+ β_0 = true mean bill length of Adelie penguins (in the population)

If species = Chinstrap, bill length = $\beta_0 + \beta_1 + \varepsilon$

β_1 = difference in mean bill length between Adelie and Chinstrap penguins

Categorical predictor

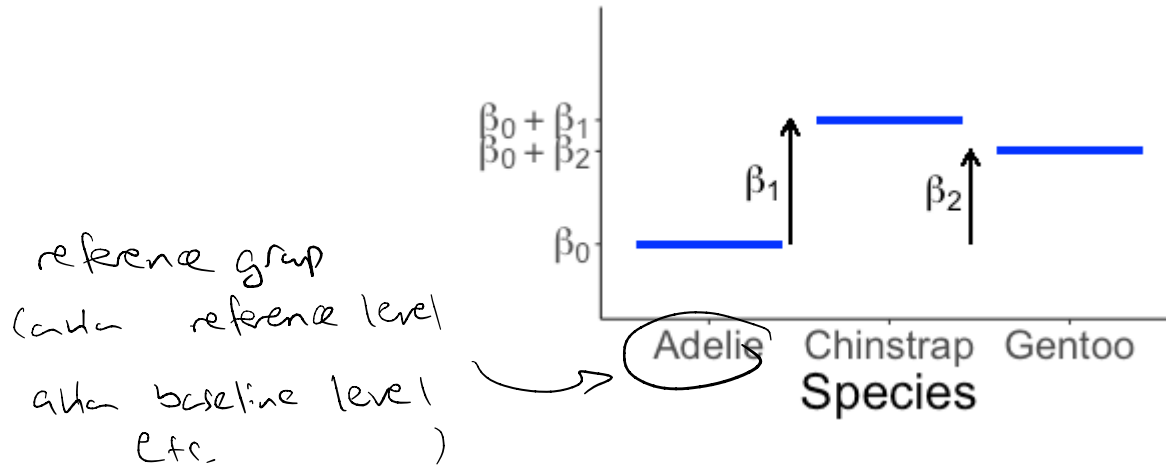


If species = Adelie, bill length = $\beta_0 + \varepsilon$

If species = Chinstrap, bill length = $\beta_0 + \beta_1 + \varepsilon$

What about Gentoo penguins?

Categorical predictor



If species = Adelie, bill length = $\beta_0 + \varepsilon$

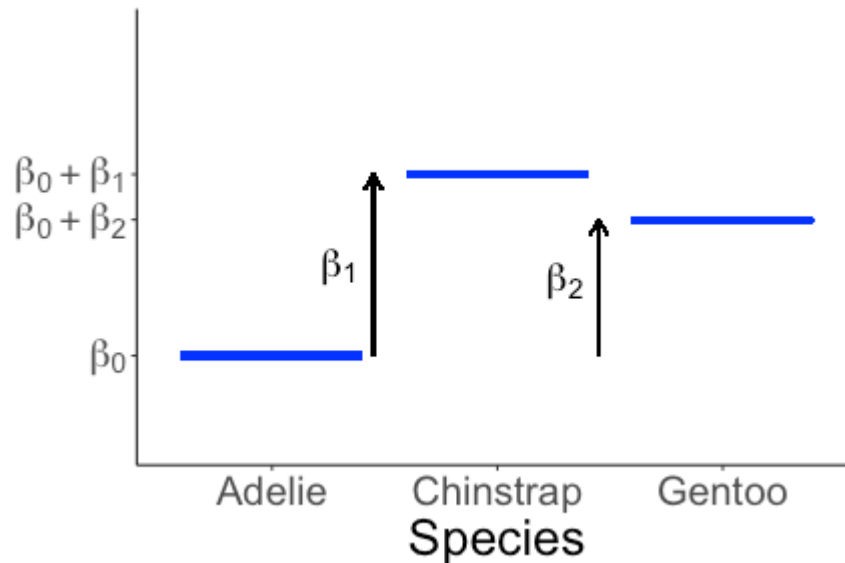
If species = Chinstrap, bill length = $\beta_0 + \beta_1 + \varepsilon$

If species = Gentoo, bill length = $\beta_0 + \beta_2 + \varepsilon$

β_2 = mean difference in bill length between Gentoo and Adelie penguins

In this model, the coefficients are relative to the Adelie penguins

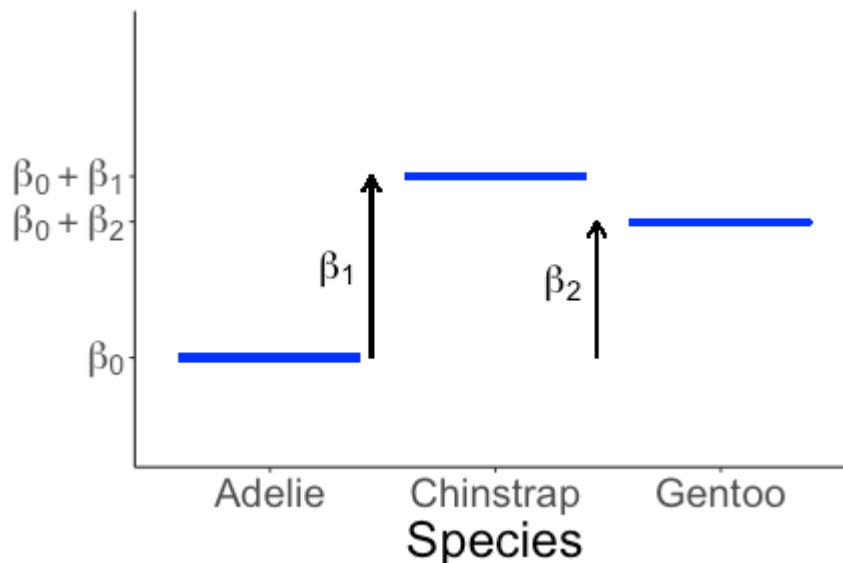
Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

(piecewise function)

Categorical predictor



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Can we write this more concisely?

Indicator variables

$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Do I need another indicator variable to represent Adelie penguins?

No: Adelie is what we get when

$$\text{IsChinstrap} = \text{IsGentoo} = 0$$

In general: represent a categorical variable that has K levels with $K-1$ indicator variables

Indicator variables

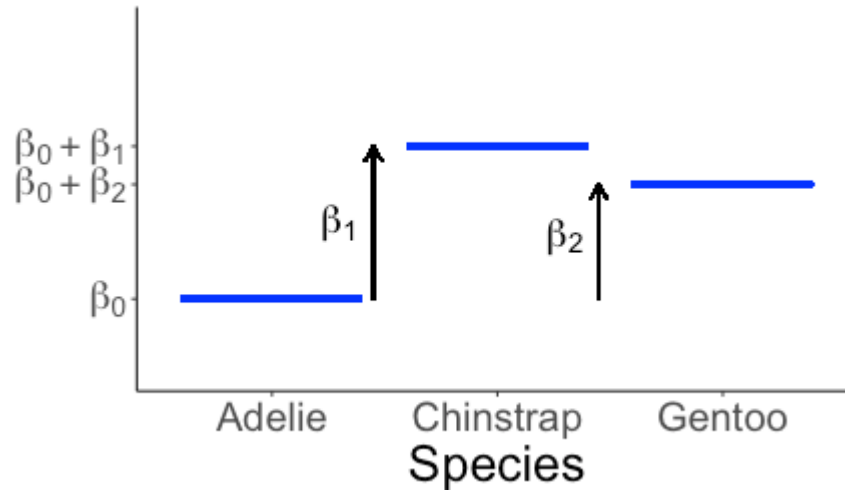
$$\text{IsChinstrap} = \begin{cases} 0 & \text{species} \neq \text{Chinstrap} \\ 1 & \text{species} = \text{Chinstrap} \end{cases}$$

$$\text{IsGentoo} = \begin{cases} 0 & \text{species} \neq \text{Gentoo} \\ 1 & \text{species} = \text{Gentoo} \end{cases}$$

Then:

Species	IsChinstrap	IsGentoo
Adelie	0	0
Chinstrap	1	0
Gentoo	0	1

Indicator variables



$$\text{bill length} = \begin{cases} \beta_0 + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Or, more concisely:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Estimated model

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

- + What is the estimated average bill length for each of the three species of penguin?
- + There are 333 penguins in the dataset used to fit the model. How many degrees of freedom does the fitted model have?

$$\begin{aligned} \text{Adelie : } \widehat{\text{bill length}} &= 38.82 + 10.01(0) + 8.74(0) \\ &= 38.82 \end{aligned}$$

$$\begin{aligned} \text{Chinstrap : } \widehat{\text{bill length}} &= 38.82 + 10.01(1) + 8.74(0) \\ &= 38.82 + 10.01 = 48.83 \end{aligned}$$

$$\begin{aligned} \text{Gentoo : } \widehat{\text{bill length}} &= 38.82 + 8.74 = 47.56 \end{aligned}$$

Estimated model

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

Species	$\widehat{\text{bill length}}$
Adelie	38.82
Chinstrap	$38.82 + 10.01 = 48.83$
Gentoo	$38.82 + 8.74 = 47.56$

Estimated model

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Species	$\widehat{\text{bill length}}$
Adelie	38.82
Chinstrap	$38.82 + 10.01 = 48.83$
Gentoo	$38.82 + 8.74 = 47.56$

Degrees of freedom: $333 - 3 = 330$

+ (Lose one degree of freedom for each parameter estimated)

$(\beta_0, \beta_1, \beta_2)$

Fitting the model in R

R will automatically create indicator variables:

(for categorical variables)

```
lm(bill_length_mm ~ species, data = penguins_no_nas)
```

```
##
```

```
## Call:
```

```
## lm(formula = bill_length_mm ~ species, data = penguins_n
```

```
##
```

```
## Coefficients:
```

```
##          (Intercept)    speciesChinstrap    speciesGentoo
```

```
##          38.824          10.010          8.744
```

$$\widehat{\text{bill length}} = 38.82 + 10.01 \text{ IsChinstrap} + 8.74 \text{ IsGentoo}$$

Class activity

- + Practice regression with a single categorical predictor, with experimental data on fruit flies
- + No need to submit this one on Canvas

https://sta112-f25.github.io/class_activities/ca_15.html